Saturday Test — No. 3

Take your time. Show your work where it helps.

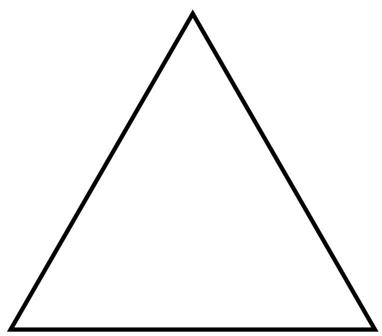
Name ____________________________ Date _____________ Time: 15 min

Numbers and Shapes

1. Fill in the missing numbers in the sequence:

3, 7, 11, _______, _______, 23

2. How many lines of symmetry does an equilateral triangle have? Draw them on the figure below.



Answer: _____________

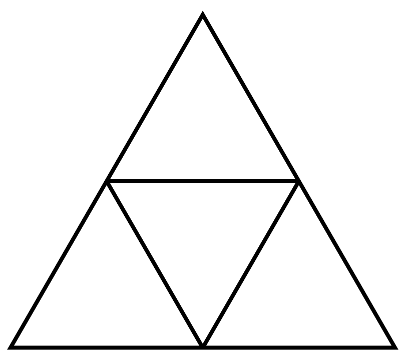
3. A rectangle has whole-number side lengths and a perimeter of 20 cm. What dimensions give the largest possible area? What is that area?

4. List every prime number between 20 and 40.

Counting Carefully

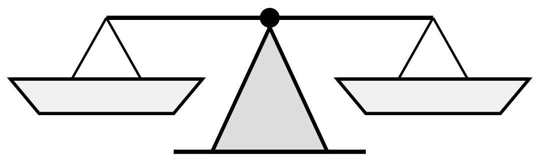
5. Write down the whole numbers from 1 to 20. How many times does the digit ‘1’ appear in your list?

6. In the figure below, the three midpoints of an equilateral triangle have been connected. How many triangles can you find in the figure? Count triangles of every size.



Answer: ____________

Reasoning



7. You have 9 coins that look identical, but one is slightly heavier than the rest. Using a balance scale, find the heavier coin in just 2 weighings. Describe your strategy.

8. Fill in the empty cells of the grid below so that every row, every column, and both diagonals add to 15. Use each of the digits 1 through 9 exactly once.

2	**9**	
	5	

Answer Key — for parents

1. 15, 19. Constant difference of 4.

2. 3 lines of symmetry. One through each vertex to the midpoint of the opposite side.

3. 5 cm × 5 cm, area = 25 cm^2. Among rectangles with a fixed perimeter, the square has the largest area. Other options: 1×9 (9), 2×8 (16), 3×7 (21), 4×6 (24).

4. 23, 29, 31, 37. (21=3×7, 25=5×5, 27=3^3, 33=3×11, 35=5×7, 39=3×13.)

5. 12. Ones place: 1, 11 contribute 2 ones. Tens place: 10–19 contribute 10 ones. Total 12. Common slip: forgetting that 11 has two 1s.

6. 5 triangles. Three small upright triangles at the corners, one small inverted triangle in the centre, and the original large triangle.

7. Split into three groups of 3. *Weighing 1:* weigh group A against group B. If they balance, the heavy coin is in group C; otherwise it is in the heavier of A or B. Take that group of 3. *Weighing 2:* weigh one coin against another, set the third aside. Same logic identifies the coin. The key insight: a balance has three outcomes, so divide into thirds, not halves.

8. Magic square solution:

2	**9**	**4**
7	**5**	**3**
6	**1**	**8**